



Saad Mughal | F2023-009
Khadijah Zahoor | F2023-956
Team no: 10
Team Name: patches
Total Marks: 100
Due 17 May 2026

HCI & Computer Graphics

Milestone 3

Initial Prototype & User Test Plans

Course Lecturer: Sania Khalid
Lab Instructor: Sania Khalid
Section: C

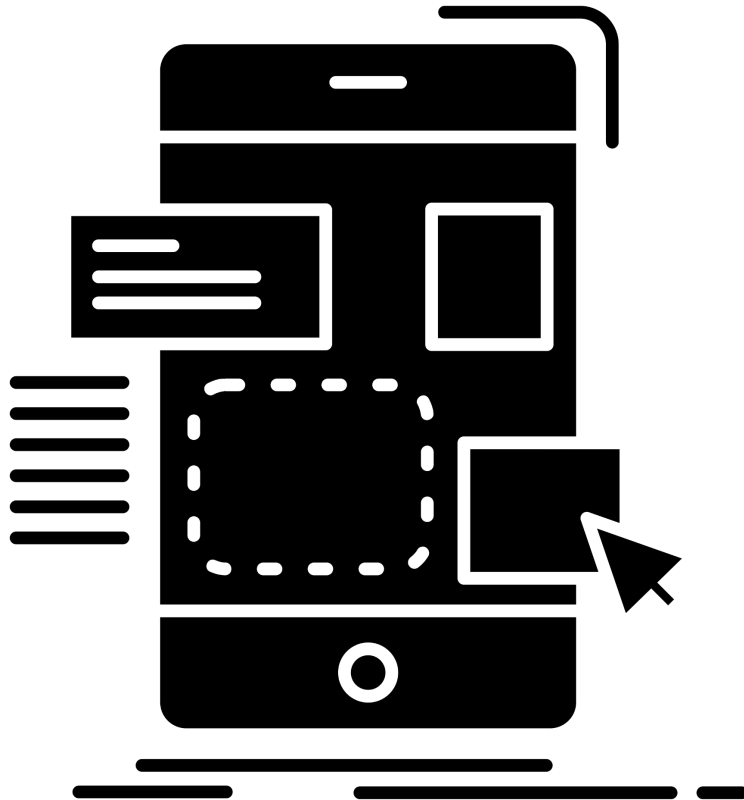


Table of Contents

Section 1: Wireframes.....	2
1.1 Key Screens.....	2
WF-01: Home Dashboard.....	2
WF-02: Scan Receipt.....	3
WF-03: Receipt Verification.....	3
WF-04: Digital Khata (Ledger).....	4
WF-05: Analytics Dashboard.....	4
WF-06: AI Budget Assistant.....	5
WF-07: Item Price History.....	5
WF-08: Receipt Detail View.....	6
WF-09: Budget Setup.....	7
WF-10: Notifications & Alerts.....	7
WF-11: Manual Entry Form.....	8
WF-12: Onboarding.....	8
1.2 Navigation Flows.....	9
Flow A: Scan & Verify a Receipt.....	9
Flow B: Check Price History for an Item.....	10
Flow C: Manual Expense Entry.....	10
Flow D: Monthly Spending Review.....	11
Flow E: Respond to a Price Alert.....	11
Section 2: Prototype.....	12
2.1 Prototype Scope - 6 Key Tasks.....	12
2.2 Dummy Data Summary.....	13
2.3 Prototype Link.....	13
Section 3: User Test Plan.....	13
3.1 User Test Plan Overview.....	14
Purpose and High-Level Goals.....	14
Recruiting Criteria and Strategy.....	14
Test Components.....	15
Sequence of Activities per Session.....	15
Task Instructions.....	15
3.2 Informed Consent Form.....	18
3.3 Logging Sheet - Critical Incident & Task Observation Form.....	19
Task Observation Log.....	20
Critical Incident Log.....	20
General Observations.....	21
3.4 User Test Script - Moderator Guide.....	21

Before the Session.....	21
Welcome and Introduction.....	21
Consent.....	22
Think-Aloud Training.....	22
Task Introduction.....	23
During Each Task.....	23
After All Tasks.....	24
Debrief Discussion.....	24
Closing.....	24
3.5 Post-Test Questionnaire - System Usability Scale (SUS).....	25
SUS Scoring Instructions (for researchers only, do not show to participants).....	26
Open-Ended Follow-Up (optional, read aloud after questionnaire is collected).....	26

Milestone 3: Initial Prototype & User Test Plans

HCI & Computer Graphics Course Project Documentation

Section 1: Wireframes

All wireframes were produced in Figma using a low-fidelity convention: single typeface (Inter), no colour other than black and white, and rough placeholder shapes to encourage high-level feedback.

1.1 Key Screens

The thirteen screens below represent the primary UI states of DailyKhata. Each screen is described with its layout intent, key elements, and the Chart of Accounts context where applicable.

WF-01: Home Dashboard

Central landing screen. Top bar shows the app name and avatar. A hero card displays the monthly total (Rs. 28,400) against the budget (Rs. 35,000) with a progress bar at 81%. Four quick-action cards below: Scan Receipt, View Khata, Ask AI, Analytics. Two recent receipt cards follow, each showing vendor, account code pill (e.g. 1000 Food & Groceries), total, and item count. A price alert banner warns of an 18% Atta price increase. Bottom nav: Home | Khata | Analytics | AI Assistant.

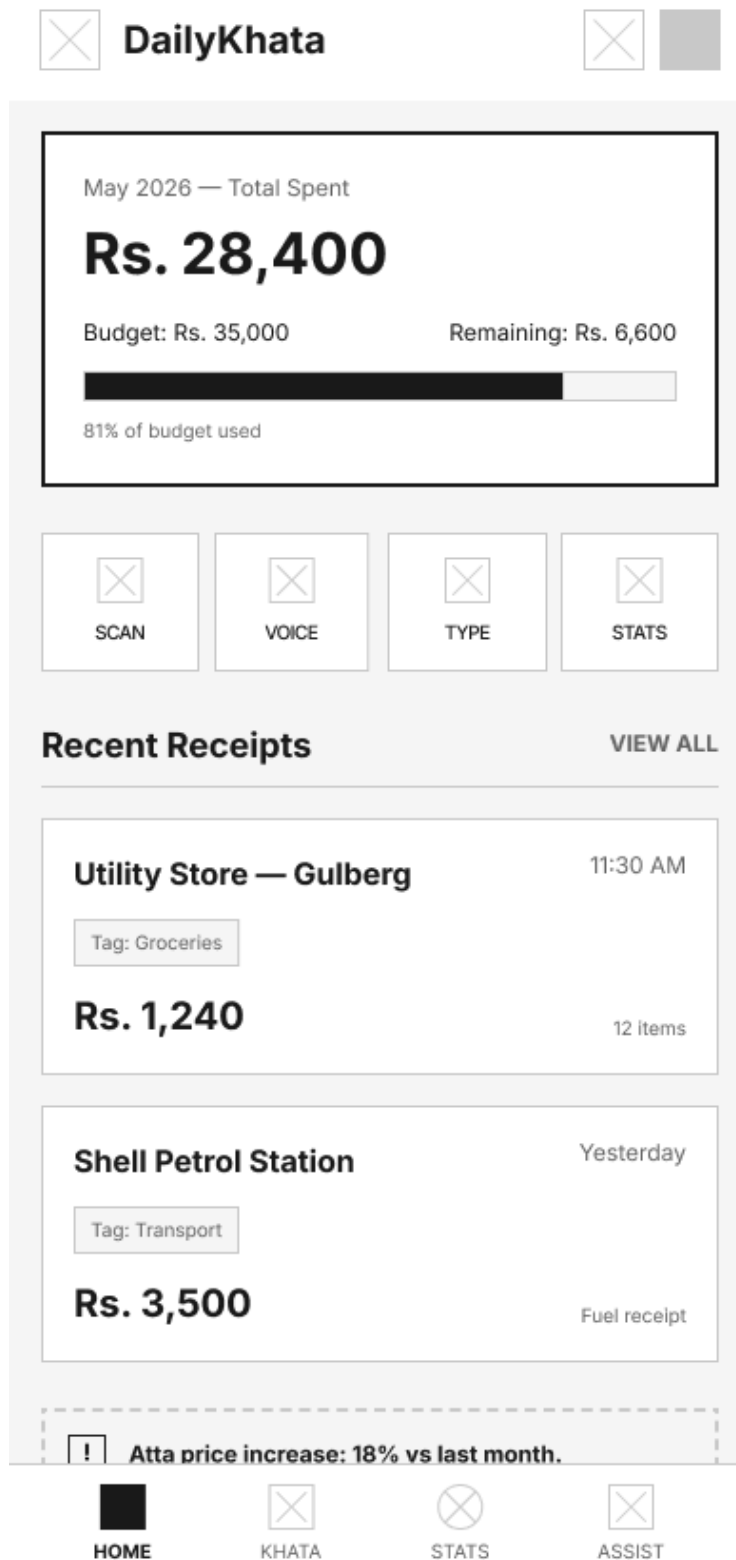


Figure WF-01: Home Dashboard - hero spend card, quick actions, recent receipts, price alert banner.

WF-02: Scan Receipt

Full-screen camera viewfinder with a dashed alignment rectangle and four corner bracket marks. A bottom strip contains three options: Capture (primary circle button), Upload from Gallery, and Enter Manually. A processing state card below the viewfinder shows 'AI Processing...' with a progress bar. Supports handwritten, printed, and market receipts.

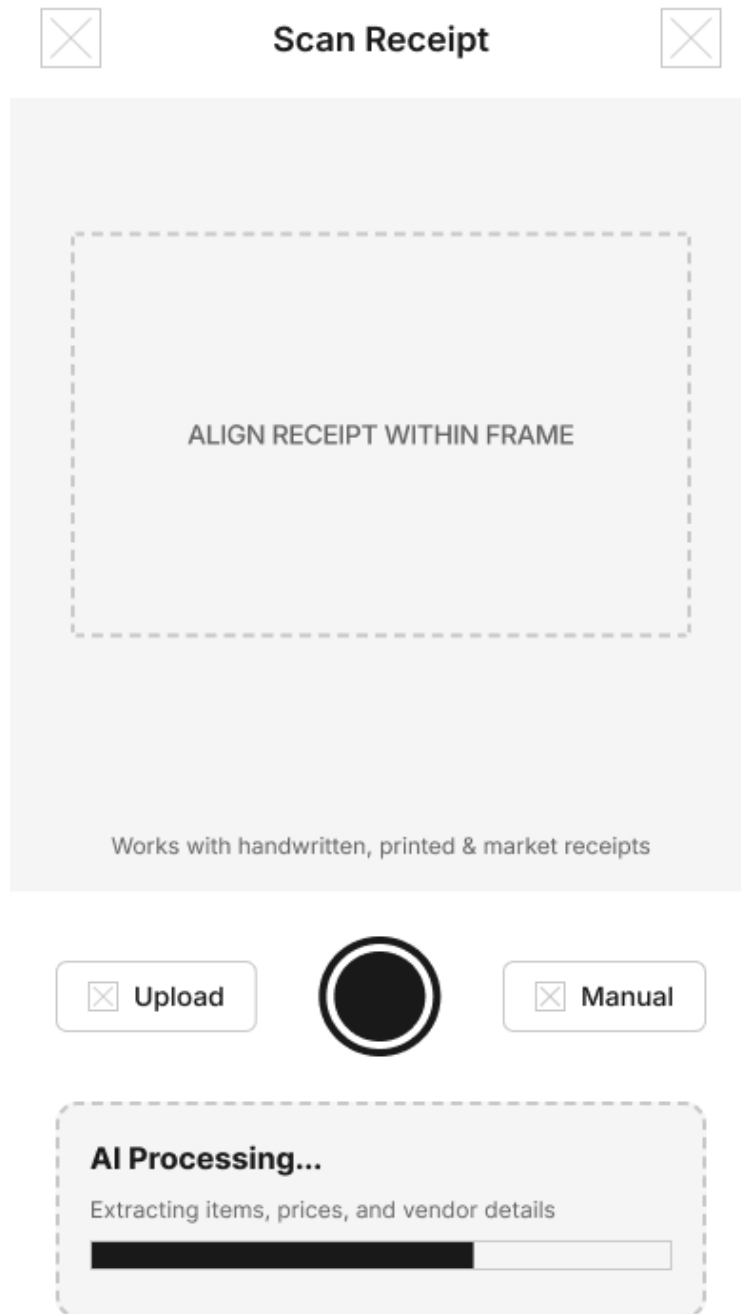


Figure WF-02: Scan Receipt - camera viewfinder, capture/upload/manual options, AI processing state.

WF-03: Receipt Verification

Post-scan verification screen. Vendor header card shows name, date/time, and account pill. An itemized table displays Item, Qty, Unit Price, and Total for each scanned line. Fields with low OCR confidence are highlighted with a yellow dashed border and a tap-to-correct warning. Below the table, a price comparison section shows month-over-month changes per item. Two action buttons: Confirm & Save to Khata (primary) and Edit Items (secondary).

Verify Receipt

Save

Utility Store — Gulberg, Lahore

Receipt Date: 16 May 2026 · 11:30 AM

1000 · Food & Groceries

ITEM	QTY	UNIT	TOTAL
Atta (10kg)	1	212	Rs. 212
Doodh (1L)	4	180	Rs. 720
Chawal (5kg)	1	340	Rs. 340
Cooking Oil	2	480	Rs. 960
Total			Rs. 2,232

3 fields flagged — tap to correct before saving

Price Changes Since Last Visit

Atta: Rs. 180 → Rs. 212 +18%

Doodh: Rs. 175 → Rs. 180 +3%

CONFIRM & SAVE TO KHATA

EDIT ITEMS

Figure WF-03: Receipt Verification - itemized table, yellow uncertainty flags, price comparison, confirm/edit.

WF-04: Digital Khata (Ledger)

Full digital ledger view. A four-card summary strip shows Spent, Budget, Saved, and total Receipts count. A search bar supports full-text search across vendors, items, and accounts. Horizontal filter pills allow filtering by account category. Entries are grouped by date, each row showing the account code pill, vendor name, amount, item count, and a scan verification badge. Bottom nav: Khata tab active.

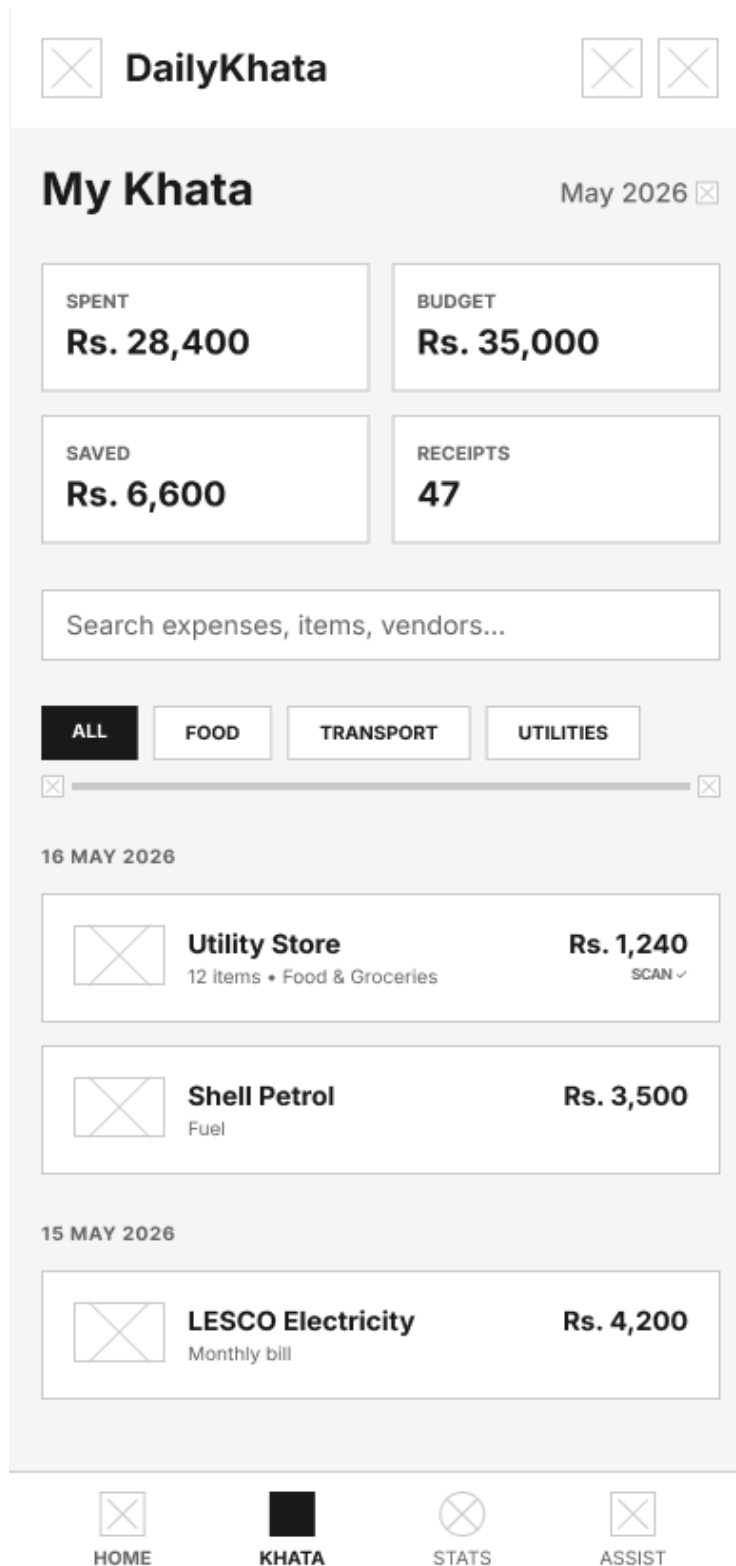


Figure WF-04: Digital Khata - date-grouped ledger, account code pills, search, filter pills.

WF-05: Analytics Dashboard

Spending analytics screen. A horizontal bar chart section shows spending by Chart of Accounts category with percentage labels. A month-over-month comparison card highlights increases (red) and decreases (green) per category. A price tracker section lists tracked items with their purchase-to-purchase price changes and a trending indicator arrow.

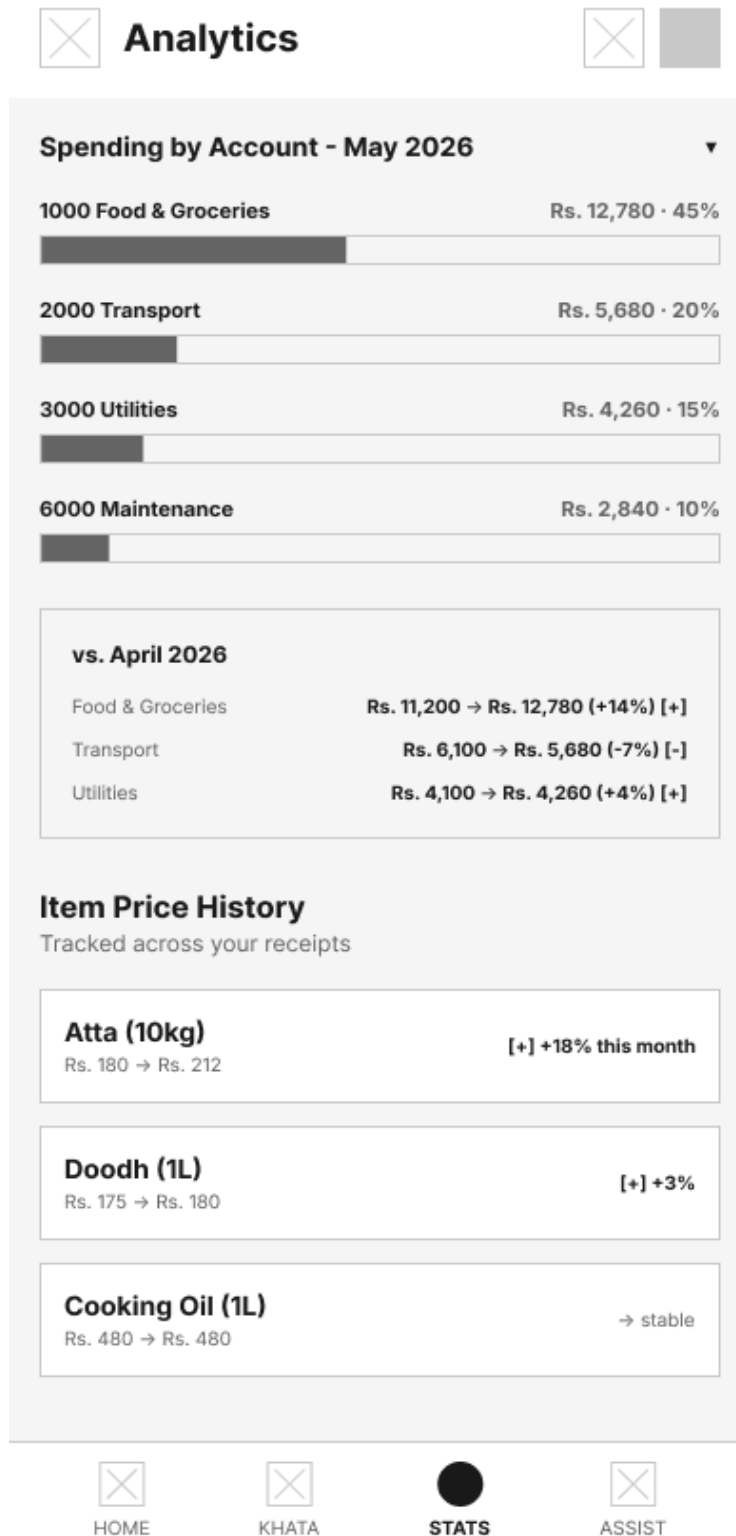


Figure WF-05: Analytics Dashboard - account-based bar chart, MoM comparison, item price tracker.

WF-06: AI Budget Assistant

Conversational AI interface. An AI badge at the top reads 'Powered by DailyKhata AI Based on your receipts.' The chat area shows alternating user (dark background) and AI (light background) message bubbles. AI responses include inline data cards with vendor breakdowns and price trend rows. A pinned input bar at the bottom contains a text field, a send button, and three quick-question pills: This month's total, Biggest expense, Price alerts.

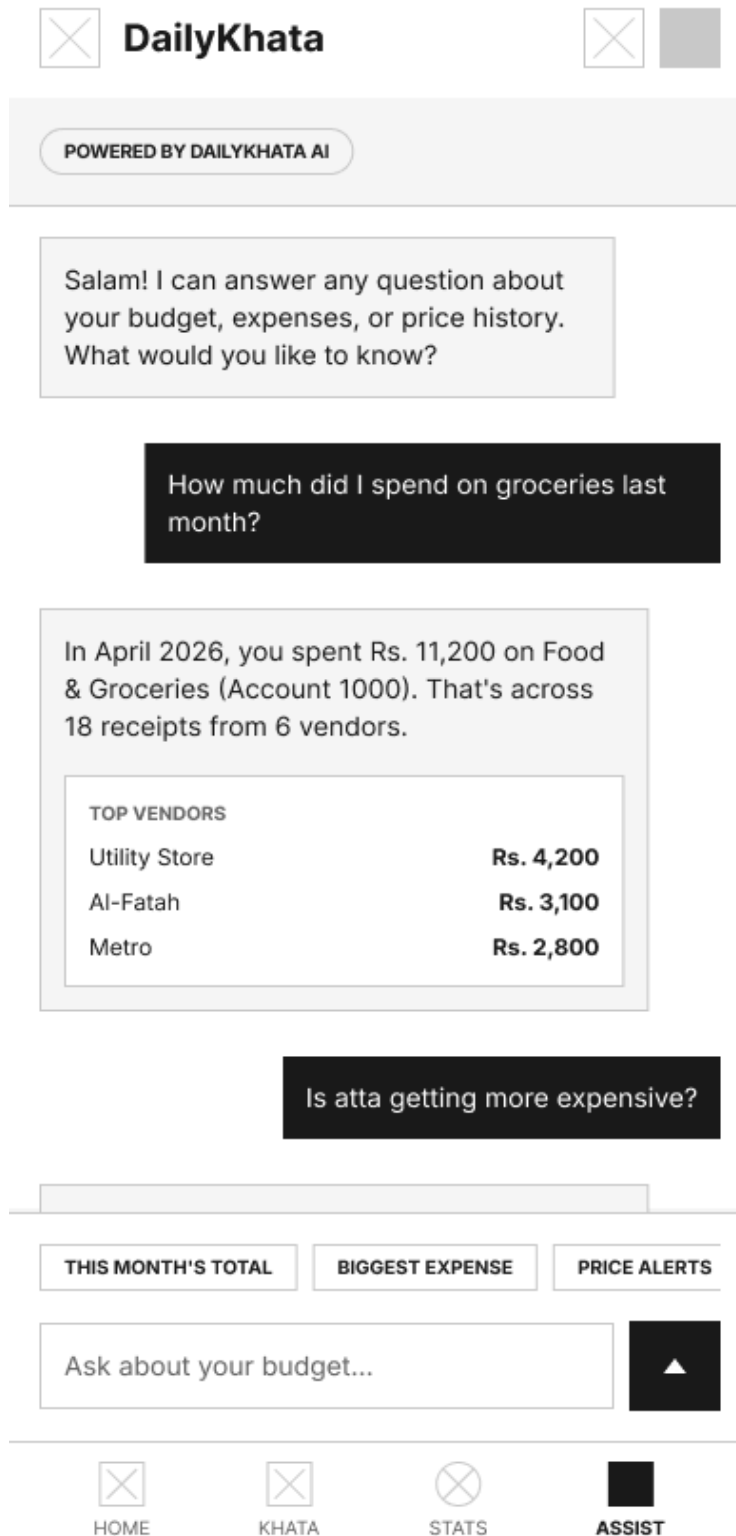


Figure WF-06: AI Budget Assistant - conversational chat, inline data cards, price trend answers, quick-question pills.

WF-07: Item Price History

Dedicated price history screen for any scanned item. A search bar at the top allows lookup by item name. The currently viewed item card (e.g. Atta 10kg) shows vendor count, receipt count, and first-seen date. A price timeline table lists Month, Vendor, Price, and Change with color-coded percentage indicators. A simple bar chart below shows price progression. An insight card at the bottom provides a plain-language recommendation based on the trend.

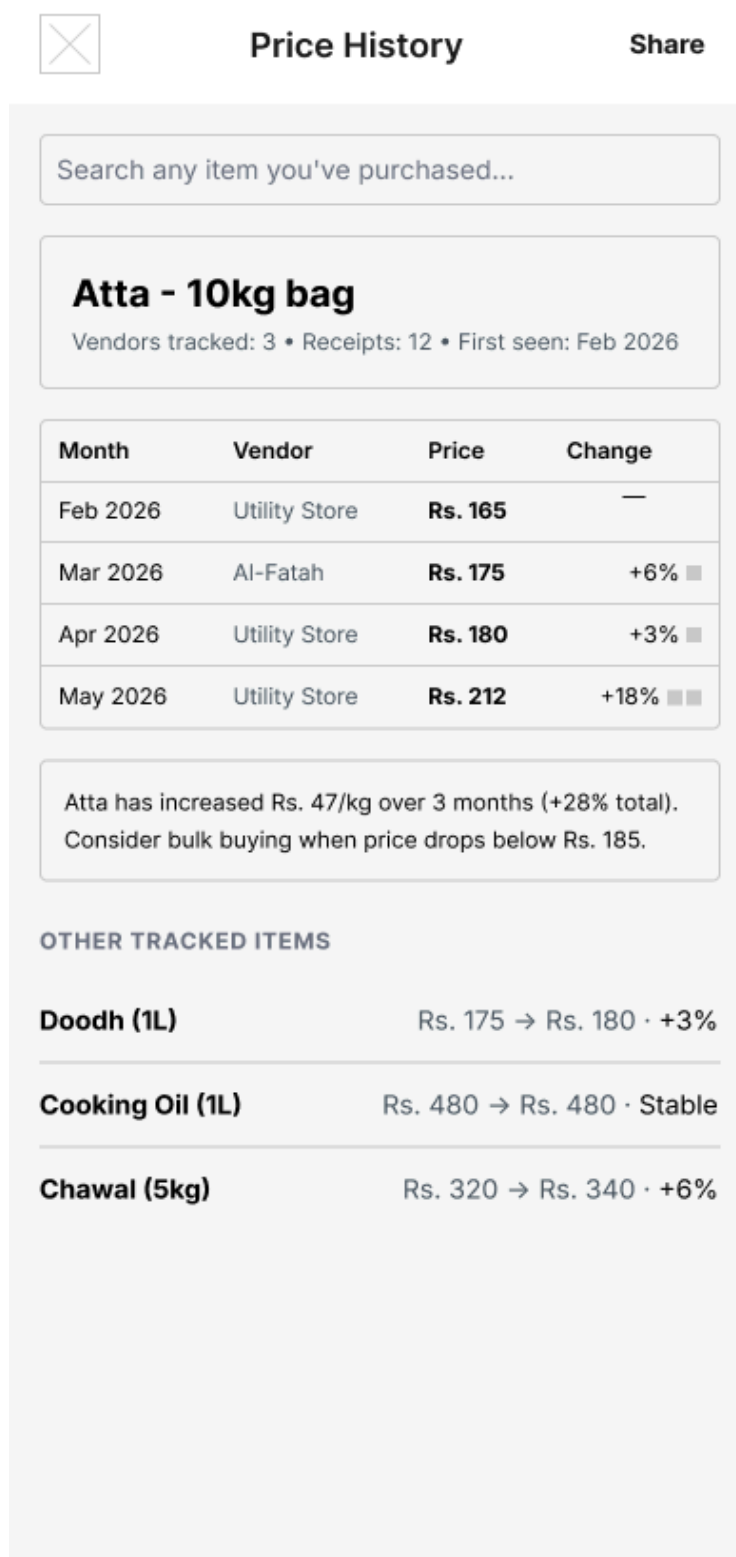


Figure WF-07: Item Price History - timeline table, bar chart, insight recommendation card.

WF-08: Receipt Detail View

Full receipt detail screen. A dark header card (navy background) shows vendor name, location, date/time, account classification, and a receipt ID. An itemized table below lists every scanned line item with Qty, Unit Price, Total, and an inline price change indicator per item. A receipt scan thumbnail placeholder appears below the table. Three action buttons at the bottom: Share Receipt, Delete, Re-categorise.



Receipt Detail

Edit

Utility Store - Gulberg

16 May 2026 • 11:30 AM

Account: 1000 • Food & Groceries

Receipt ID: #RCP-2026-0847

ITEM	QTY	UNIT	TOTAL
Atta (10kg) ▲ +18% vs last purchase	1	212	Rs. 212
Doodh (1L) ▲ +3% vs last purchase	4	180	Rs. 720
Chawal (5kg) ▲ +6% vs last purchase	1	340	Rs. 340
Cooking Oil → Same vs last purchase	2	480	Rs. 960
Subtotal			Rs. 2,232

[Original receipt scan]

SHARE RECEIPT

DELETE

RE-CATEGORISE

Figure WF-08: Receipt Detail View - dark vendor card, itemized table with price deltas, scan thumbnail, actions.

WF-09: Budget Setup

Budget configuration screen. A monthly income card at the top accepts the user's income figure with a type toggle (Salary, Freelance, Both). Below, each Chart of Accounts category has its own budget row with an editable amount input and a live percentage-of-income label. A read-only 'Remaining / Savings' row at the bottom auto-calculates. A validation row shows total allocated vs savings in green.

Budget Settings

Save

MONTHLY INCOME

Rs. 75,000

Salary

Freelance

Both

Allocate Budget by Account

Following Chart of Accounts

1000 Food & Groceries

Rs. 18,000

24%

2000 Transport

Rs. 7,500

10%

4000 Healthcare

Rs. 3,750

5%

5000 Education

Rs. 5,250

7%

6000 Maintenance

Rs. 3,000

4%

Remaining / Savings

Rs. 31,500

42%

Total allocated: Rs. 43,500 • Savings: Rs. 31,500

Figure WF-09: Budget Setup - income input, Chart of Accounts budget allocation rows, live savings calculation.

WF-10: Notifications & Alerts

Alerts center. Filter pills at the top: All, Price Alerts, Budget Alerts, Receipts. Unread alerts show a colored left-border accent (orange for price alerts, red for budget warnings). Each notification row shows an icon, bold title, description with context (vendor, amount, percentage), and a timestamp. Read alerts appear without the accent border.

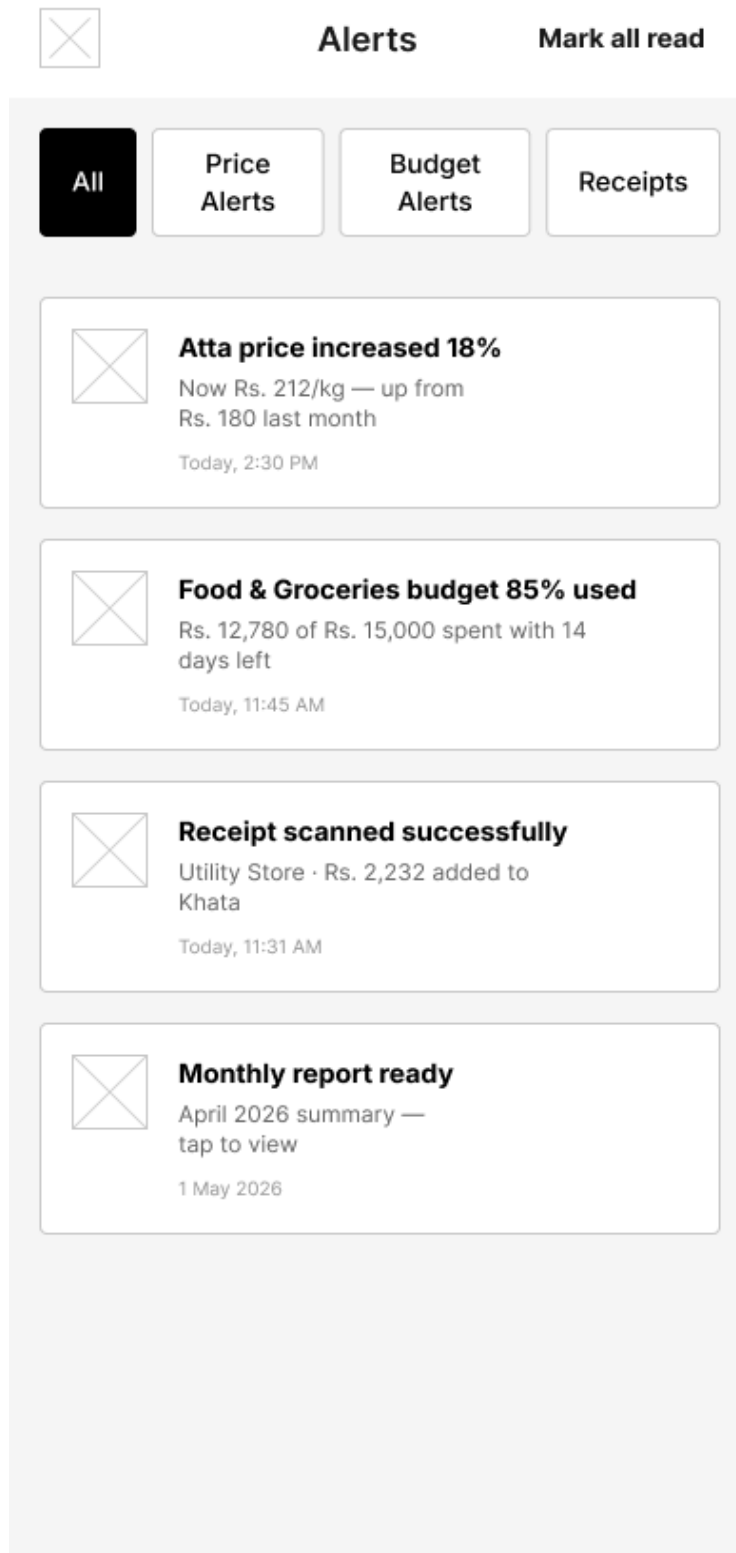


Figure WF-10: Notifications & Alerts - colored alert borders, price and budget alerts, filter pills.

WF-11: Manual Entry Form

Manual expense entry form. Tab strip at the top: Receipt Scan | Manual Entry (active). Form fields: Vendor Name, Date (with calendar picker icon), Account (Chart of Accounts dropdown), Amount, and Note. Below a divider, an optional itemized section allows adding individual line items with name, quantity, and unit price. A calculated total updates live. A primary Save to Khata button sits at the bottom.



Manual Entry

Vendor Name

e.g. Local dairy, Ahmed Kiryana

Date

16 May 2026

Account

1000 · Food & Groceries



Amount

Rs. ____

Description / Note

e.g. weekly ration, fuel refill

ITEMS (OPTIONAL)

Atta	1	212	Rs. 212
Doodh	4	180	Rs. 720

+ Add Item

Entry Total

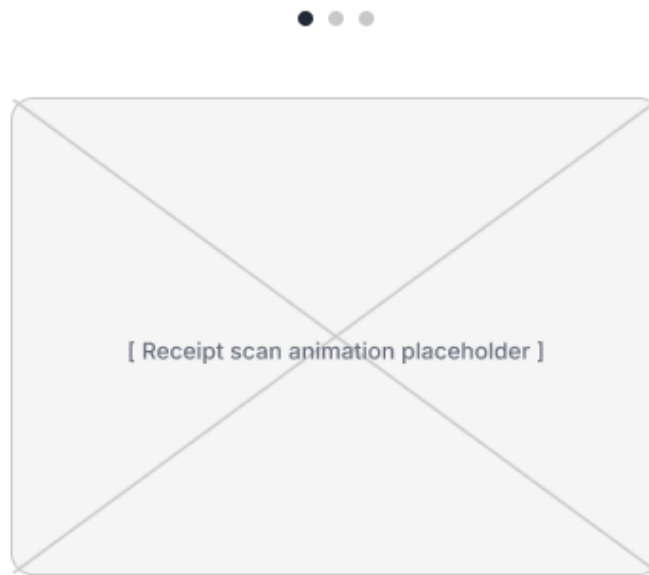
Rs. 932

SAVE TO KHATA

Figure WF-11: Manual Entry Form - Chart of Accounts dropdown, optional itemization, live total.

WF-12: Onboarding

Full-screen onboarding slide (screen 1 of 3). Progress dots at the top. A large illustration placeholder for a receipt scan animation. Bold heading: 'Your receipts. Digitized instantly.' with a supporting subheading. Three feature pills: Auto digitize, Price tracking, AI insights. Primary Get Started button (no account required). A sign-in link and offline capability note complete the screen.



Your receipts. Digitized instantly.

Scan any receipt — handwritten, printed,
or market — and DailyKhata automatically
builds your complete expense history.

☒ Auto digitize

☒ Price tracking

☒ AI insights

Get Started

I already have an account — [Sign in](#)

No manual entry required · Works offline

Figure WF-12: Onboarding screen 1 - animation placeholder, value proposition, feature pills, CTA.

WF-13: Voice Entry

Full-screen voice logging interface. A language toggle (Urdu / EN) sits top-right. A status badge reads "Listening - AI is transcribing in real time." A large pulsing microphone animation occupies the center with a live waveform and timer below. A Live Transcript box shows the spoken input in real time. An AI Detected section lists parsed items with amounts as the user speaks. Two action buttons at the bottom: Re-record and Confirm & Save.

Voice Entry

Urdu • EN

LISTENING - AI IS TRANSCRIBING IN REAL TIME

Tap mic to pause · 00:08

LIVE TRANSCRIPT

"Aaj maine atta liya do sau bara ka, doodh ek liter aur ek pao chai patti..."

AI DETECTED · 3 ITEMS

Atta 1 bag	Rs. 212
Doodh 1 liter	Rs. 180
Chai patti 1 pao	Rs. 320

Re-record

Confirm & Save

Figure WF-13: Figure WF-13: Voice Entry - live transcript, AI-detected items, waveform, Re-record / Confirm & Save.

1.2 Navigation Flows

Five key navigation flows are documented below. Each flow corresponds directly to a core task in the prototype and the user test. Insert miniaturised wireframe images side-by-side with arrows showing screen transitions.

Flow A: Scan & Verify a Receipt

User taps Scan Receipt on the home dashboard, captures a receipt via camera or gallery, reviews the AI-parsed itemized table (correcting any yellow-flagged fields), confirms, and lands on the Receipt Detail view showing the saved entry.

Step	Screen	Trigger to Next Screen
1	WF-01 Home	Tap Scan Receipt
2	WF-02 Scan Receipt	Capture / Upload
3	WF-03 Receipt Verification	Confirm & Save
4	WF-08 Receipt Detail	End of flow

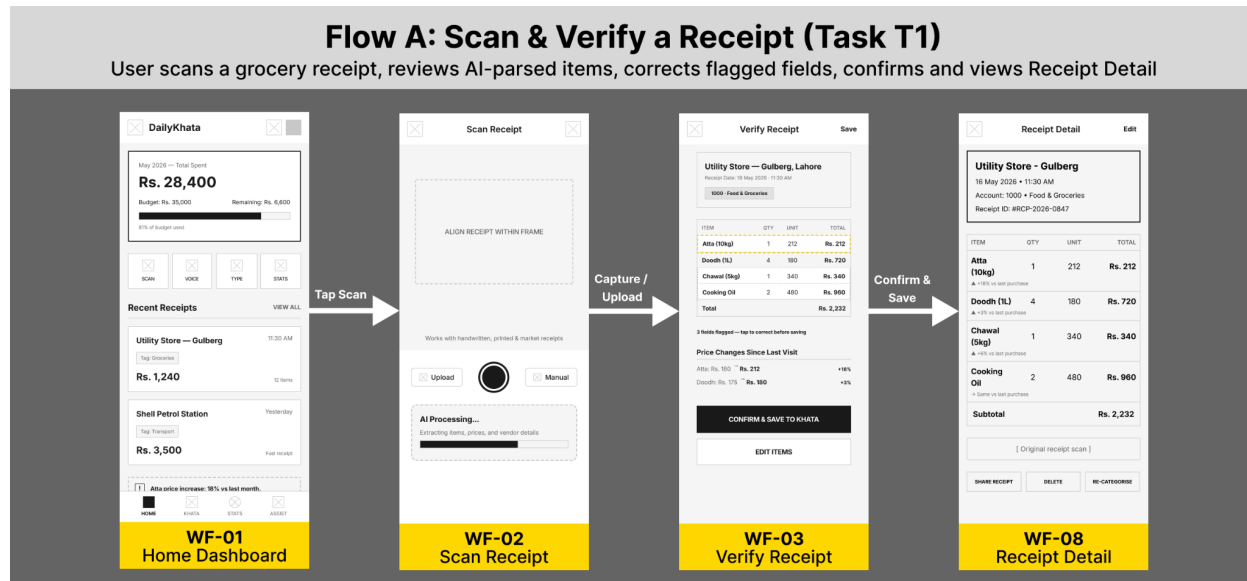


Figure Flow A: Scan & Verify a Receipt navigation flow with screen transitions.

Flow B: Check Price History for an Item

User opens the Analytics dashboard, taps a tracked item in the Price Tracker section to view the full price timeline and bar chart, then optionally navigates to the AI Assistant to ask a natural-language question about the item's trend.

Step	Screen	Trigger to Next Screen
1	WF-05 Analytics	Tap item in Price Tracker
2	WF-07 Price History	Ask AI for context
3	WF-06 AI Assistant	End of flow

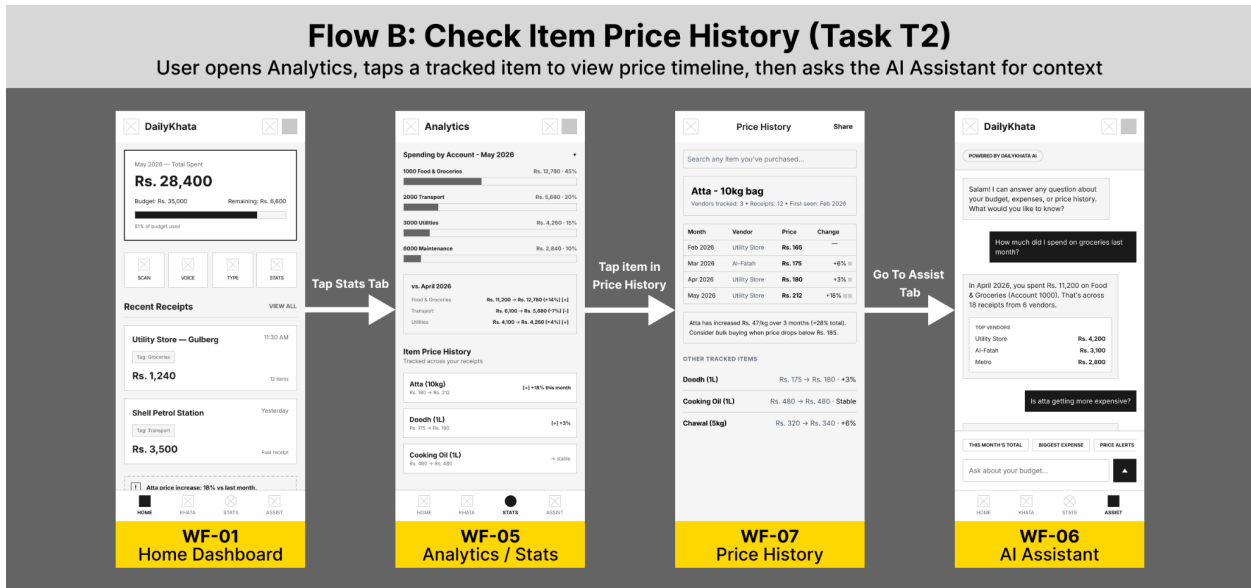


Figure Flow B: Check Price History for an Item navigation flow with screen transitions.

Flow C: Manual Expense Entry

User taps the Enter Manually option from the Scan Receipt screen, fills in the Chart of Accounts dropdown, amount, and optional itemization, saves, and the entry appears in the Digital Khata ledger.

Step	Screen	Trigger to Next Screen
1	WF-01 Home	Tap Manual (Scan screen)
2	WF-11 Manual Entry	Save to Khata
3	WF-04 Khata Ledger	End of flow

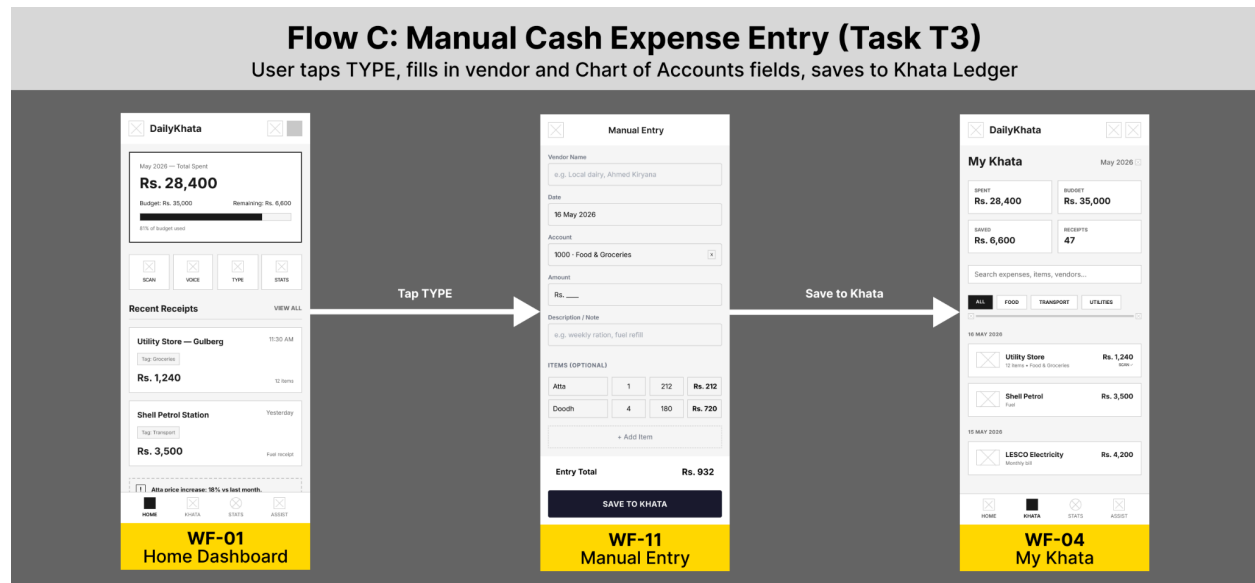


Figure Flow C: Manual Expense Entry navigation flow with screen transitions.

Flow D: Monthly Spending Review

User reviews this month's entries in the Khata ledger, switches to the Analytics dashboard to view account-level breakdowns and month-over-month changes, then navigates to Budget Setup to adjust allocations based on observed patterns.

Step	Screen	Trigger to Next Screen
1	WF-01 Home	Tap View Khata
2	WF-04 Khata Ledger	Tap Analytics tab
4	WF-09 Budget Setup	End of flow

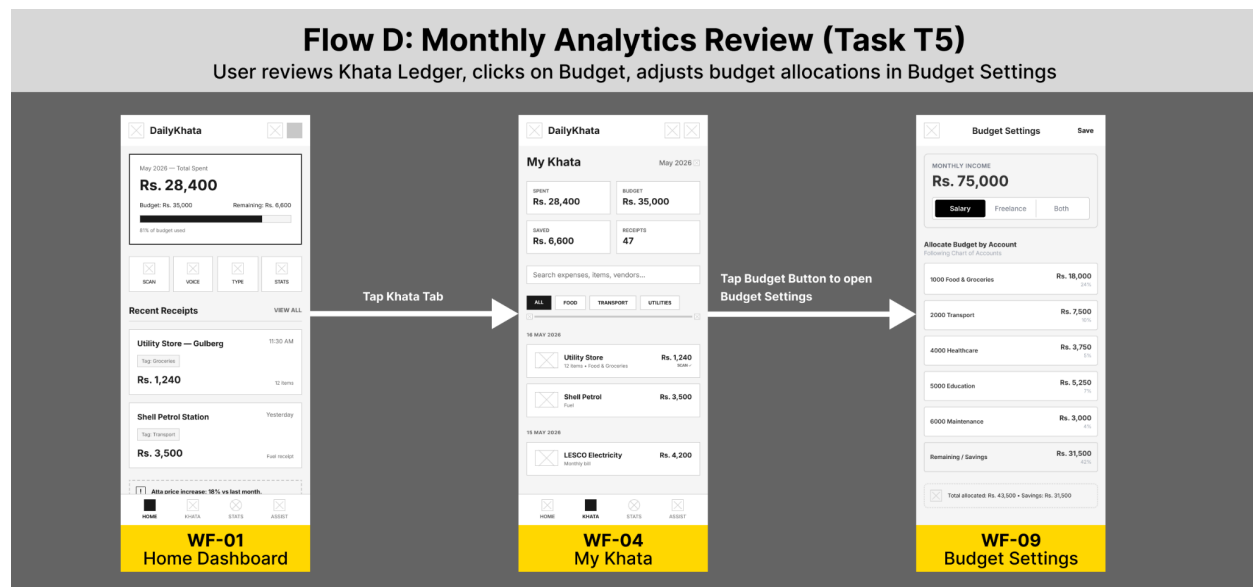


Figure Flow D: Monthly Spending Review navigation flow with screen transitions.

Flow E: Respond to a Price Alert

The user receives a price alert notification, taps it to open the Item Price History screen for the affected item, reviews the price timeline and insight recommendation, then asks the AI Assistant for a budget adjustment suggestion.

Step	Screen	Trigger to Next Screen
1	WF-10 Alerts	Tap price alert
2	WF-07 Price History	Ask AI for advice
3	WF-06 AI Assistant	End of flow

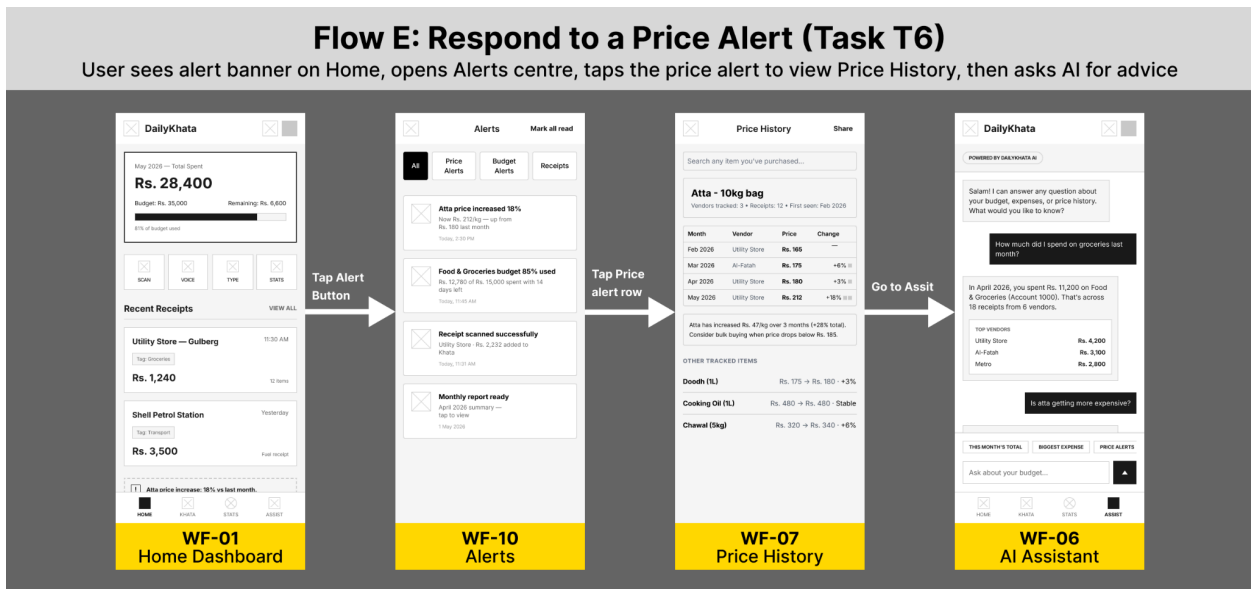


Figure Flow E: Respond to a Price Alert navigation flow with screen transitions.

Section 2: Prototype

The low-fidelity prototype is built in Figma using prototype connections between frames to create a fully clickable, task-based experience. It covers all six tasks defined for the user test and includes screens for likely error paths and wrong interactions. All dynamic elements use realistic Pakistani-context dummy data.

2.1 Prototype Scope - 6 Key Tasks

The six tasks below were selected to cover receipt scanning with AI verification, price history lookup, manual entry, AI assistant interaction, ledger review, and budget management. Together they exercise every major branch of the app.

#	Task Name	Persona	Primary Flow	Key Screens
T1	Scan and verify a grocery receipt	Rukhsana (homemaker)	Flow A	WF-01, WF-02, WF-03, WF-08
T2	Check if Atta price has increased over the last 3 months	Rukhsana	Flow B	WF-05, WF-07
T3	Add a cash expense manually (plumber visit)	Rukhsana	Flow C	WF-01, WF-11, WF-04
T4	Ask the AI how much was spent on groceries last month	Zaid (student)	Flow B variant	WF-06
T5	Review May analytics and compare to April	Zaid	Flow D	WF-04, WF-05, WF-09
T6	Act on a price alert for Cooking Oil	Zaid	Flow E	WF-10, WF-07, WF-06

2.2 Dummy Data Summary

All prototype frames use the following dummy data.

Account	Sample Entries	Amounts (Rs.)	Notes
1000 Food & Groceries	Utility Store - Atta, Doodh, Chawal, Oil	2,232 per receipt	Verification screen T1
2000 Transport	Shell Petrol, Rickshaw, Bus fare	3,500 / 120 / 50	Various receipts
3000 Utilities	LESCO Electricity, SNGPL Gas	4,200 / 2,800	Monthly bills
6000 Maintenance	Plumber visit (cash)	800	Manual entry T3
Price: Atta 10kg	Feb Rs.165, Mar Rs.175, Apr Rs.180, May Rs.212	+28% Total	Price history T2
Price: Cooking Oil 1L	Apr Rs.460, May Rs.510	0.11	Alert flow T6
Monthly Total	All categories combined	Rs. 28,400	Analytics T5
Budget	Income Rs.75,000 - allocated Rs.43,500	Savings Rs.31,500	Budget setup T5

2.3 Prototype Link

Publicly accessible Figma prototype link below.

<https://www.figma.com/proto/dTuQ4yL1cYNPl4BRI9wzqZ/DailyKhata-%7C-M03-Wireframes?node-id=1-3&t=vkKpHnfXsqE4l04e-1>

Section 3: User Test Plan

This section contains the complete user test plan for the first DailyKhata usability evaluation. It comprises five sub-documents: (3.1) Test Plan Overview, (3.2) Informed Consent Form, (3.3) Logging Sheet, (3.4) User Test Script, and (3.5) Post-Test Questionnaire (SUS).

3.1 User Test Plan Overview

Purpose and High-Level Goals

This usability test evaluates the DailyKhata low-fidelity prototype with representative users from both target groups. The test is exploratory and formative: its primary aim is to identify usability problems before the high-fidelity build, not to produce statistically significant benchmarks.

- Determine whether users can scan and verify a receipt through the AI pipeline without assistance.
- Assess the clarity of the itemized verification table and yellow-flagged uncertainty indicators.
- Evaluate whether the price history and month-over-month analytics are understandable without a tutorial.
- Test the discoverability and utility of the conversational AI assistant for budget queries.
- Identify navigation breakdowns, Chart of Accounts labelling confusion, and friction points across all six tasks.
- Collect subjective satisfaction scores using the System Usability Scale (SUS).

Recruiting Criteria and Strategy

Participants will be recruited using convenience and snowball sampling from the existing pool used in Milestone 2, supplemented by additional contacts where needed.

Criterion	Group A - Low Digital Literacy	Group B - High Digital Literacy
Age range	28 to 50	18 to 28
Occupation	Homemaker / household budget manager	University student / freelancer
Device	Android smartphone (mid-range)	Android smartphone (any)
Expense tracking	Notebook or mental tracking	Has tried or currently uses a digital app
Language	Urdu-primary, basic English	Bilingual Urdu / English
Number of participants	2 to 3	2 to 3

Total target	5 participants minimum	
--------------	------------------------	--

Exclusion criteria: Participants who are current DailyKhata team members or who have seen the prototype prior to the session are excluded. All participants must actively manage a personal or household budget and must receive physical receipts regularly.

Test Components

- Pre-test background questionnaire (2 minutes): demographics and current expense tracking habits.
- Task-based prototype walkthrough: 6 tasks, think-aloud protocol (25 to 35 minutes).
- Post-test SUS questionnaire (5 minutes).
- Optional debrief discussion: 5 minutes, open-ended.

Sequence of Activities per Session

Step	Activity	Duration	Who
1	Welcome participant, introduce study, obtain signed consent	5 min	Moderator
2	Administer pre-test background questionnaire	2 min	Participant
3	Explain think-aloud protocol, run practice task	3 min	Moderator + Participant
4	T1: Scan and verify a grocery receipt	5 min max	Participant
5	T2: Check Atta price history (3 months)	5 min max	Participant
6	T3: Add a cash expense manually	4 min max	Participant
7	T4: Ask the AI about last month's grocery spend	4 min max	Participant
8	T5: Review May analytics vs April	5 min max	Participant
9	T6: Act on a Cooking Oil price alert	5 min max	Participant
10	Post-test SUS questionnaire	5 min	Participant
11	Debrief: open discussion, final comments	5 min	Moderator
12	Thank participant, end session	1 min	Moderator

Task Instructions

The following instructions will be read aloud verbatim at the start of each task. Moderators must not provide hints or guidance during a task unless the participant is stuck for more than 90 seconds, at which point a neutral prompt ('What would you try next?') is permitted once.

T1: Scan and Verify a Grocery Receipt

Instruction to read aloud:

Imagine you have just returned from the Utility Store in Gulberg and you have a grocery receipt for your weekly shopping. Please use DailyKhata to scan this receipt and save it to your Khata. The receipt is placed in front of you.

Key observations to record:

- Does the user find the Scan Receipt option without hesitation?
- Does the user understand the yellow dashed fields and know to tap them to correct?
- Does the user read the price comparison section before confirming?
- Does the user complete the confirm step or exit early?
- Time from task start to confirmation screen.

T2: Check Atta Price History

Instruction to read aloud:

You want to know whether the price of Atta has been going up over the past few months based on your past receipts. Please use the app to find this information.

Key observations to record:

- Does the user navigate to Analytics or search directly?
- Is the Price Tracker section discovered without prompting?
- Does the user understand the price timeline table and bar chart?
- Does the user read the insight recommendation card?

T3: Add a Cash Expense Manually

Instruction to read aloud:

You paid a plumber Rs. 800 in cash yesterday and did not receive a receipt. Please add this expense to your Khata manually.

Key observations to record:

- Does the user find the manual entry path (via Scan screen or directly)?
- Is the Chart of Accounts dropdown intuitive for selecting Household Maintenance?
- Does the user complete all fields or skip the itemization section?
- Does the saved entry appear in the Khata ledger as expected?

T4: Ask the AI About Last Month's Grocery Spend

Instruction to read aloud:

You want to find out exactly how much you spent on groceries last month. Please ask DailyKhata's AI assistant this question.

Key observations to record:

- Does the user discover the AI Assistant tab without prompting?
- Is the natural-language question phrased freely or does the user look for a menu?
- Does the user understand the AI's response and the inline vendor breakdown card?
- Does the user attempt a follow-up question?

T5: Review May Analytics and Compare to April

Instruction to read aloud:

It is the end of May. Please use the app to understand how your spending this month compares to last month across your main expense categories.

Key observations to record:

- Does the user navigate to the Analytics tab directly?
- Is the month-over-month comparison card understood without explanation?
- Does the user try to drill down into any specific category?
- Does the user interact with the budget setup from the analytics screen?

T6: Act on a Cooking Oil Price Alert

Instruction to read aloud:

You received a notification that the price of Cooking Oil has increased. Please find this alert and use the app to understand the price change and decide what to do about it.

Key observations to record:

- Does the user find the Alerts screen from the bell icon or bottom nav?
- Does the user tap the alert and navigate to the Price History screen correctly?
- Does the user read the insight recommendation card?
- Does the user ask the AI for advice or attempt to adjust the budget?

3.2 Informed Consent Form

Please read this form carefully. Ask questions if anything is unclear before signing.

DailyKhata Usability Study - Informed Consent Form

Purpose of the Study

You are being invited to participate in a usability study conducted by Saad Mughal and Khadijah Zahoor as part of the HCI & Computer Graphics course at Beaconhouse National University (BNU). The purpose is to evaluate a prototype of DailyKhata, an AI-powered expense tracking app, to identify usability issues and improve the design.

What You Will Be Asked to Do

You will be asked to complete up to six short tasks using a prototype of the app on a device provided by the research team. The session will take approximately 45 to 50 minutes. You will be asked to think aloud, narrating your actions and thoughts as you work through each task.

Recording

With your permission, the session will be audio- and/or screen-recorded. Recordings will be used solely for research analysis by the project team and will not be shared externally. All recordings will be deleted within 30 days of the course submission deadline.

Confidentiality

All data collected will be anonymised. Your name will not appear in any report or presentation. You will be referred to by a participant code (for example, P01). No personally identifiable information will be published or shared outside the project team.

Voluntary Participation

Your participation is entirely voluntary. You may stop the session at any time and for any reason, with no consequences. You may also choose to skip any task or question without being required to explain why.

Risks and Benefits

There are no known risks associated with participation in this study. Your input will directly help improve the design of a mobile app intended to help Pakistani households and individuals manage daily expenses more effectively.

Questions

If you have any questions about this study, please contact Saad Mughal (F2023-009) or Khadijah Zahoor (F2023-956) via BNU email or in person at BNU Section C.

Consent Declaration

- ☐ I have read and understood the information provided above.
- ☐ I voluntarily agree to participate in this usability study.
- ☐ I agree to the session being audio and/or screen-recorded (cross out if not applicable).
- ☐ I understand that I may withdraw at any time without penalty or explanation.

Participant Name:

Participant Signature:

Date:

Researcher Name:

Researcher Signature:

3.3 Logging Sheet - Critical Incident & Task Observation Form

Complete one logging sheet per participant per session. Record task success or failure, time taken, and any noteworthy observations. A critical incident is any observable event that meaningfully helps or hinders the user's ability to complete a task.

Participant ID:		Session Date:	
Moderator:		Observer:	
Device used:		Recording?	Yes / No

Task Observation Log

Task	Success? (Y/N/P)	Time (mm:ss)	Errors / Wrong Paths	Critical Incidents & Observations
T1: Scan & verify receipt				
T2: Atta price history				
T3: Manual cash entry				
T4: Ask AI about groceries				
T5: May vs April analytics				
T6: Act on price alert				

P = Partial success (task completed with significant difficulty or a moderator neutral prompt). Time is recorded from when the task instruction is read to when the task is completed or abandoned.

Critical Incident Log

Record each critical incident as a separate row. Severity: 1 = minor confusion, 3 = significant delay, 5 = task failure or complete breakdown.

#	Task	Time	Description of Incident	Positive / Negative	Severity (1-5)
1					
2					
3					
4					
5					
6					
7					
8					

General Observations

Use the space below for any observations not captured in the tables above:

3.4 User Test Script - Moderator Guide

This script provides the exact wording and sequence of steps for each test session. Text in [square brackets] is a stage direction for the moderator and should not be read aloud. Indented italic text in bordered boxes is spoken verbatim.

Before the Session

[Prepare the device: open the Figma prototype to the Home screen (WF-01).]

[Ensure the logging sheet, consent form, and task cards are printed and ready.]

[Set up screen recording on a second device or use Figma's built-in recorder.]

[Place a sample handwritten grocery receipt on the table for T1.]

Welcome and Introduction

Thank you for coming in today. My name is [moderator name] and I am a student at BNU working on a class project. My colleague [observer name] will be taking notes during our session.

Today I am going to ask you to try out an early prototype of an app called DailyKhata. It is designed to help people track daily expenses by scanning receipts and using AI to answer budget questions. This is a very early version, so some things may not work perfectly, and that is completely fine.

I want to be very clear: we are testing the app, not you. There are no right or wrong answers. If something is confusing, that tells us something important about the design, not about you.

The session will take about 45 to 50 minutes. Do you have any questions before we begin?

Consent

[Hand the participant the consent form.]

Please take a moment to read this form. Let me know if you have any questions and please sign at the bottom if you agree to participate.

[Wait for signed consent. Check the recording permission box.]

[Start screen recording if consent is granted.]

Think-Aloud Training

Before we start the tasks, I would like to show you a technique called thinking aloud. While you are using the app, I would like you to narrate what you are doing and what you are thinking. For example: 'I am looking for a button to scan a receipt' or 'I am not sure what this icon means.'

Let's try a quick practice. Can you open the calculator on this device and narrate what you are doing as you multiply 1234 by 2?

[Observe. If the participant goes silent, prompt: 'Can you tell me what you are thinking right now?']

Great. That is exactly the kind of narration I am looking for during the tasks.

Task Introduction

I am going to give you a series of short tasks to complete in the app. I will read each task aloud and there will also be a printed task card in front of you. Please attempt each task as you normally would.

I cannot answer questions about how the app works during the tasks, but please keep narrating your thoughts out loud.

[Place the device in front of the participant with the Home screen (WF-01) visible.]

During Each Task

[Read the task instruction aloud from Section 3.1. Place the printed task card in front of the participant.]

[Start timing. Take notes on the logging sheet.]

[If the participant is silent for more than 15 seconds, prompt: 'What are you thinking right now?']

*[If the participant is stuck for more than 90 seconds, prompt once: 'What would you try next?'
Do not provide the answer.]*

*[If the participant cannot complete the task after 3 minutes, say: 'Let us move on to the next task.'
Record as failure (N) on the logging sheet.]*

[Between tasks, reset the prototype to the correct starting screen.]

After All Tasks

Thank you, that is all the tasks. You did a great job.

I am going to give you a short questionnaire now. It has ten statements and I would like you to rate how much you agree or disagree with each one based on your experience with the app today. There are no right or wrong answers.

[Hand the participant the SUS questionnaire. Allow up to 5 minutes.]

Debrief Discussion

Now that you have filled in the questionnaire, I would love to hear any final thoughts. Is there anything about the app that you found particularly easy, confusing, or surprising?

Was there anything you expected the app to do that it did not do?

If you could change one thing about the app, what would it be?

[Record responses. Let the participant lead. Do not prompt further beyond these three questions.]

Closing

That is everything. Thank you so much for your time today. Your feedback will directly shape the next version of this app. Is there anything else you would like to add?

[Stop recording. Collect the signed consent form and completed SUS questionnaire.]

[Thank the participant and escort them out.]

3.5 Post-Test Questionnaire - System Usability Scale (SUS)

The System Usability Scale (SUS) is a validated ten-item Likert questionnaire that produces a single usability score between 0 and 100. It is used here because it is quick (under 5 minutes), tool-agnostic, and provides a benchmark score that can be compared across prototype iterations. Scores above 68 are considered above average; scores above 80 are considered excellent.

DailyKhata - Post-Test Questionnaire (SUS)

Participant ID: _____ Date: _____ Session #: _____

For each statement below, please circle or tick one number to indicate how strongly you agree or disagree based on your experience with DailyKhata today. 1 = Strongly Disagree, 5 = Strongly Agree.

#	Statement	1 Strongly Disagree	2	3	4	5 Strongly Agree
1	I think that I would like to use this app frequently.					
2	I found the app unnecessarily complex.					
3	I thought the app was easy to use.					
4	I think that I would need the support of a technical person to use this app.					
5	I found the various functions in this app were well integrated.					
6	I thought there was too much inconsistency in this app.					
7	I would imagine that most people would learn to use this app very quickly.					
8	I found the app very cumbersome to use.					
9	I felt very confident using the app.					
10	I needed to learn a lot of things before I could get going with this app.					

SUS Scoring Instructions (for researchers only, do not show to participants)

- Odd-numbered items (1, 3, 5, 7, 9): converted score = participant rating minus 1.
- Even-numbered items (2, 4, 6, 8, 10): converted score = 5 minus participant rating.
- Sum all ten converted scores and multiply by 2.5 to get the final SUS score (0 to 100).

Raw Score Sum: _____	SUS Score (x 2.5): _____	Grade / Band: _____
-----------------------------	---------------------------------	----------------------------

Open-Ended Follow-Up (optional, read aloud after questionnaire is collected)

Q: What was the easiest part of using DailyKhata today?

Q: What was the most confusing or frustrating part?

Q: Is there anything you wished the app could do that it currently does not?
